

Two-Channel ASEP with Narrow Entrances

Reproduction of Pronina & Kolomeisky (2007) + GPU acceleration

Your Name

August 16, 2026

Outline

- 1 Model & Reproduction
- 2 MFT vs MC (professor's focus)
- 3 Spontaneous Symmetry Breaking
- 4 Computational Optimization
- 5 Conclusions & Outlook

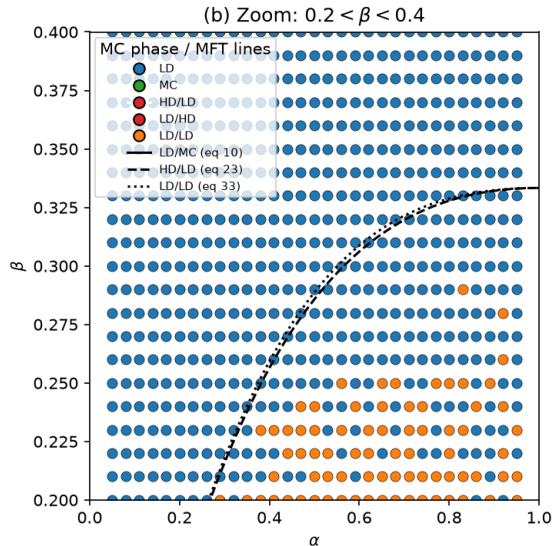
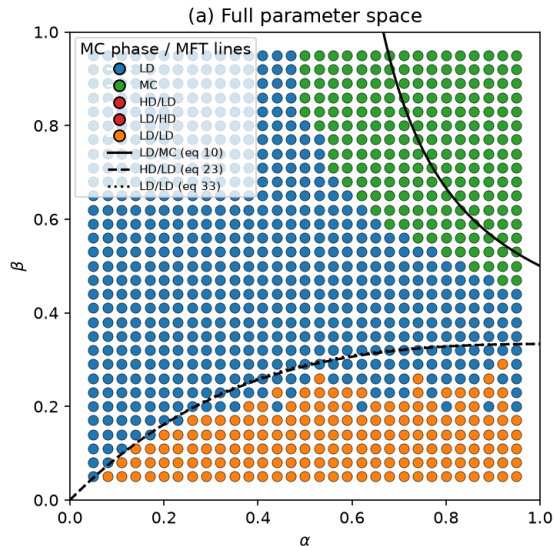
The Model

Two parallel channels, hard-core exclusion:

- Channel 1: hop right (rate 1)
- Channel 2: hop left (rate 1)
- **Narrow-entrance coupling:** enter channel 1 only if exit site of channel 2 is empty (and vice versa)
- Exit rate β independent of other channel

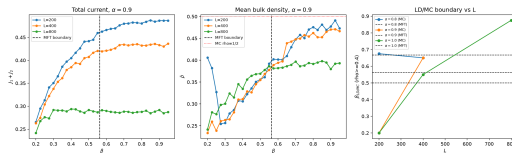
Reproduced exactly: phase diagram, currents, densities, and $P(\rho_1, \rho_2)$.

Phase Diagram (Fig 2)



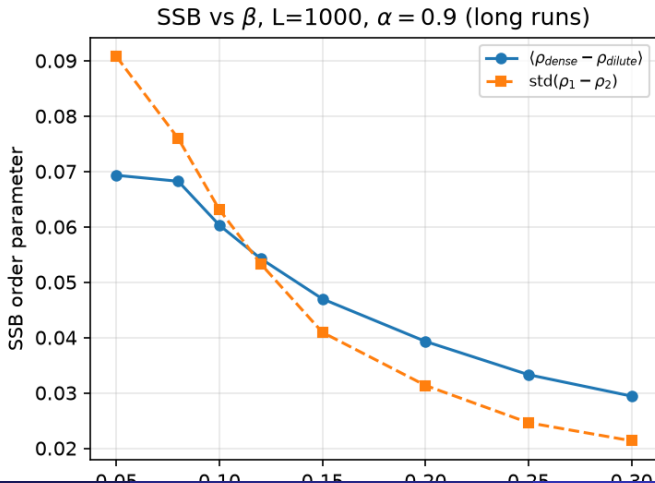
MFT transition positions are NOT exact

- Unlike single-lane TASEP (exact at $\alpha = \beta = 1/2$), here MFT deviates from MC.
- MC LD/MC boundary sits at **higher** β than MFT.
- Deviation grows with α : $+0.01$ at $\alpha = 0.8 \rightarrow +0.075$ at $\alpha = 1.0$.
- Cause: MFT neglects correlations; narrow-entrance coupling creates boundary correlations (an effective boundary impurity) MFT misses.

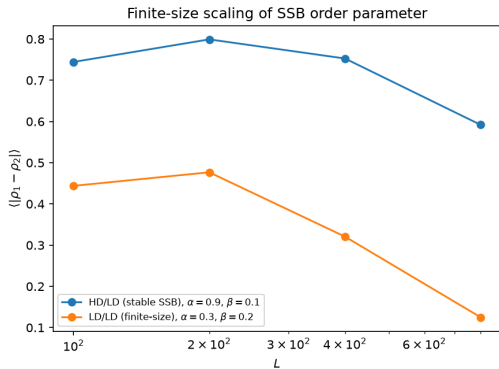


SSB: order parameter

Two asymmetric phases: HD/LD (first order), LD/LD (continuous). Robust order parameter is $\text{std}(\rho_1 - \rho_2)$ over an ensemble (the time-averaged $|\rho_1 - \rho_2|$ is washed out by state flipping).



Finite-size effect (L small vs large)

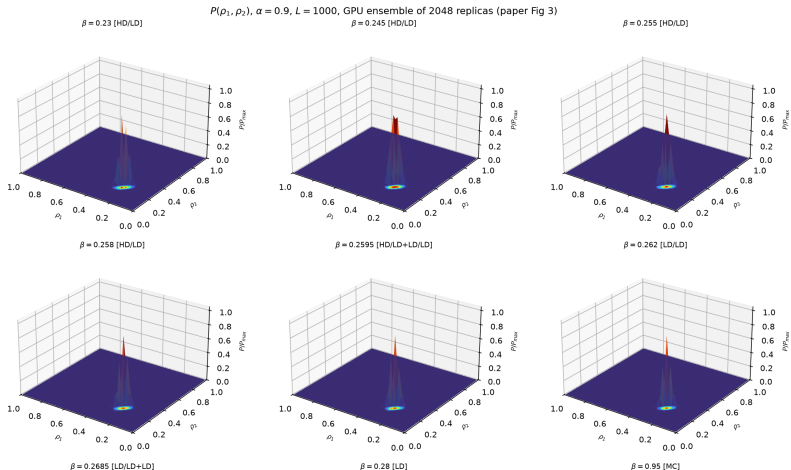


SSB order parameter vs L

At $L = 200$: strong, immediate HD/LD (dense 0.9, dilute 0.04). At $L = 1000$: broken states flip; the time-average washes out unless runs are long. The instantaneous separation persists.

$P(\rho_1, \rho_2)$ from an ensemble (Fig 3)

A single short run at $L = 1000$ stays in ONE broken state (one peak). The paper's TWO peaks are recovered by an **ensemble** of independent replicas.



BKL: classic vs Fenwick

- Classic BKL: selection via double loop over the active list, cost $\propto n_{active}$.
- Fenwick-tree BKL: selection in $O(\log L)$, total rate maintained incrementally. **4 $\times$ faster, identical physics.**

Kernel	Msteps/s
classic BKL	1.2
Fenwick BKL	4.8

GPU ensemble: the main strength

The GPU maps the ASEP poorly as a single serial chain. The strength is an **ensemble**: one CUDA thread per independent replica.

$$\text{GPU: } \underbrace{\text{rep}_1 \oplus \text{rep}_2 \oplus \cdots \oplus \text{rep}_N}_{\text{one thread each}}$$

Backend	Msteps/s
CPU, 1 thread	4.4
CPU, 12 cores	29.6
GPU ensemble (2060 SUPER)	~85

19× single-thread CPU, 2.9× 12-core CPU.

One run → all figures

- `run_all_gpu.py`: one GPU ensemble scan produces ALL MC data (fig2, fig3, fig6, SSB), saved to `results/gpu/*.npz`.
- `plot_all.py`: plots every figure by READING the saved data; no MC at plot time.

This avoids re-running MC for each figure.

Summary

- Full reproduction of the two-channel ASEP with narrow entrances.
- Confirmed the paper's message: MFT transition positions deviate from MC due to neglected boundary correlations.
- SSB characterized with a robust order parameter $\text{std}(\rho_1 - \rho_2)$ and an ensemble $P(\rho_1, \rho_2)$; strong finite-size dependence.
- GPU ensemble gives $19\times$ speedup and enables long runs / thousands of replicas.

Open questions

- Does the LD/LD band survive at $L = 1000$ and move with L ?
- Does the MFT-vs-MC deviation shrink with L or persist (genuine MF failure)? Needs longer runs at $L = 800$.
- Going beyond 2nd order / adding correlation parameters might improve MFT.